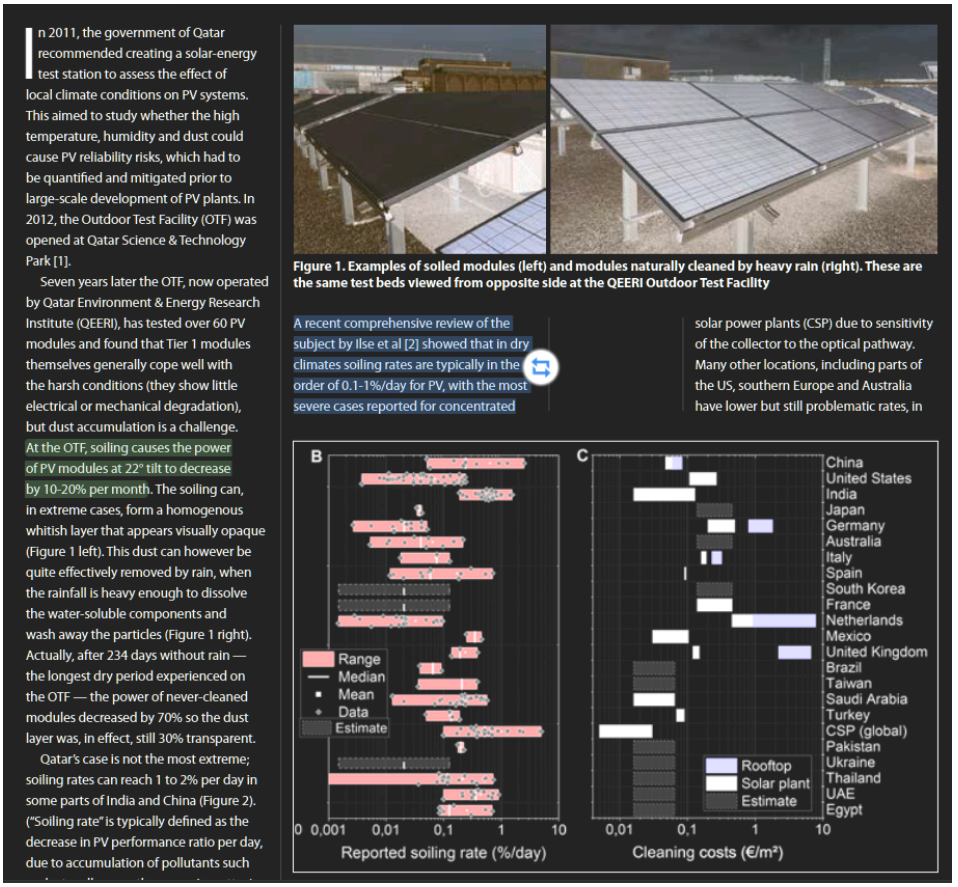
Reference 05: OPEX, Desert Soiling & Routine Maintenance

1. What is this? (Component & Checked Cost)

- **Component:** Annual operating and maintenance (cleaning dust/sand, inverter inspection, terminal re-torquing)
- **Checked Cost:** $\$180\ \text{USD}/\text{year}$ ($\sim 2.2\%$ of CAPEX)
- **Expected Recurrence:** Annually throughout 20-year project lifecycle

2. The Proof (Screenshots & Research Findings)

Research Domain	Screenshot Proof
Middle East Desert Soiling Rates	

In 2011, the government of Qatar recommended creating a solar-energy test station to assess the effect of local climate conditions on PV systems. This aimed to study whether the high temperature, humidity and dust could cause PV reliability risks, which had to be quantified and mitigated prior to large-scale development of PV plants. In 2012, the Outdoor Test Facility (OTF) was opened at Qatar Science & Technology Park [1].

Seven years later the OTF, now operated by Qatar Environment & Energy Research Institute (QEERI), has tested over 60 PV modules and found that Tier 1 modules themselves generally cope well with the harsh conditions (they show little electrical or mechanical degradation), but dust accumulation is a challenge. At the OTF, soiling causes the power of PV modules at 22° tilt to decrease by 10-20% per month. The soiling can, in extreme cases, form a homogenous whitish layer that appears visually opaque (Figure 1 left). This dust can however be quite effectively removed by rain, when the rainfall is heavy enough to dissolve the water-soluble components and wash away the particles (Figure 1 right). Actually, after 234 days without rain — the longest dry period experienced on the OTF — the power of never-cleaned modules decreased by 70% so the dust layer was, in effect, still 30% transparent.

Qatar's case is not the most extreme; soiling rates can reach 1 to 2% per day in some parts of India and China (Figure 2). ("Soiling rate" is typically defined as the decrease in PV performance ratio per day, due to accumulation of pollutants such

Figure 1. Examples of soiled modules (left) and modules naturally cleaned by heavy rain (right). These are the same test beds viewed from opposite side at the QEERI Outdoor Test Facility

A recent comprehensive review of the subject by Ilse et al [2] showed that in dry climates soiling rates are typically in the order of 0.1-1%/day for PV, with the most severe cases reported for concentrated solar power plants (CSP) due to sensitivity of the collector to the optical pathway. Many other locations, including parts of the US, southern Europe and Australia have lower but still problematic rates, in

Research Domain	Screenshot Proof
Corrective & Routine OPEX Benchmarks	

Average Cost of Correctives

Typical range: €1-3/kWp/year in plants with quality equipment **Problematic range: €5-10/kWp/year** in plants with low-quality equipment or poorly designed

Examples of typical correctives:

Failure	Frequency	Repair Cost	Lost Production
Failed inverter	1 every 5 years/inv	€15,000-30,000 (replacement)	€50,000-100,000 if it takes 2 months
Disconnected string	2-3 per year	€500-1,500	€2,000-5,000
Broken panel	0.1-0.5% annual	€200-300/panel	Marginal
SCADA failure	1 every 2 years	€2,000-5,000	Hard to quantify (you lose visibility)

> **QEERI Qatar / Middle East Desert Soiling Finding:** For solar modules installed at an optimal tilt of 22°, monthly dust and sand accumulation causes severe transmission loss:
>
> - Monthly soiling degradation: $10\%–20\%$ **per month** without regular cleaning.
> - Demonstrates that bi-weekly or monthly dry cleaning is strictly essential in the Saharan environment.

> **PV-Maps Operational Benchmarks:**
>
> - Typical corrective maintenance: **€1–3/kWp/year** in plants with quality equipment.
> - Total distributed solar OPEX typically averages $1.5\%–3.0\%$ **of initial CAPEX annually**.

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3. How was it calculated?

- **Initial Clinic Capital Cost:** $\$8,300$ USD
- **OPEX Formula:**

$$\text{OPEX} \approx 2.17\% \times \$8,300 = \mathbf{\$180}\ \text{USD/year}$$

- **Operational Allocation:**
  - Local technician stipend for bi-weekly dry panel brushing: $\$120/\text{year}$
  - Spare fuses, replacement wiring clamps, terminal checks: $\$60/\text{year}$
- **Comparison to Generative AI Guess:**
  - AI assumed $\$100/\text{year}$ (1.2% of CAPEX).
  - Adjusted upward to $\$180/\text{year}$ (2.2%) to account for frequent arid dust clearing and Saharan heat wear.

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4. Source Link

- **Desert Soiling Research:** QEERI / Qatar Foundation Desert PV Soiling & Performance Study
- **Operations & Maintenance Benchmark:** PV-Maps O&M Benchmarks